

Experimental VLE, consistency and fitting

After midterm · Session 2 of 6 · 180 minutes

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Session outcome

Separate vapor-pressure inputs, consistency evidence and fitted-model error.

Before class: Bring T , P , x_1 , y_1 data or use the explicitly synthetic example; obtain Antoine equation forms, units and validity ranges.

Learning path and all labs

The 180-minute class

IN CLASS	MINUTES
Recall and prediction	10
Concepts and derivation	45
Worked example	30
Break	10
Instructor lab demonstration	25
Guided student exploration	35
Discussion and interpretation	15
Exit question and independent task	10

The measurement-to-model chain

Measured T , P , x and y → pure-component vapor pressures → inferred activity coefficients → consistency diagnostics → regression → prediction within a stated domain.

Each arrow adds assumptions. A fitted curve does not remove errors introduced earlier.

Activities inferred from VLE

Under the ideal-vapor approximation with neglected Poynting correction,

$$\gamma_i = \frac{y_i P}{x_i P_i^{sat}(T)}$$

The absent component at a pure endpoint has no finite measured γ from this ratio. Small x or y amplifies measurement error.

Antoine inputs are part of the model

A common form is

$$\log_{10} P^{sat} = A - \frac{B}{T + C}$$

A, B and C belong to a specific equation, logarithm base, T unit and pressure unit. Lab 08 supports several forms; changing the selection does not convert constants.

Check P^{sat} at one measured temperature and verify the source range.

A vapor-pressure error becomes an activity error

Synthetic point: $P=100$ kPa, $x_1=0.4$, $y_1=0.6$ and $P_1^{\text{sat}}=120$ kPa.

$$\gamma_1 = \frac{0.6(100)}{0.4(120)} = 1.25$$

Using 150 kPa instead yields $\gamma_1=1.00$. The measured mixture data did not change; the inferred liquid behavior did.

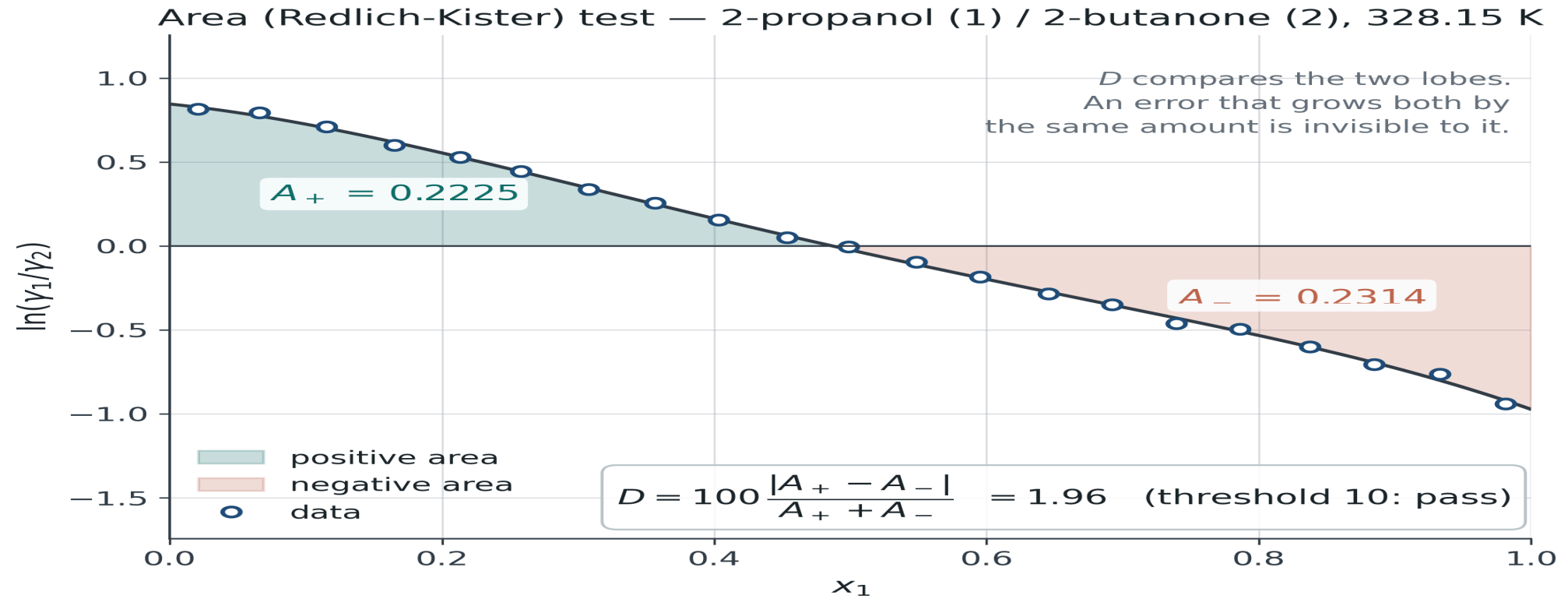
Gibbs–Duhem constrains activities

At constant T and P for a binary liquid,

$$x_1 d \ln \gamma_1 + x_2 d \ln \gamma_2 = 0$$

A VLE path may change P or T. Pressure/excess-volume and temperature/excess-enthalpy terms need appropriate treatment or an explicit approximation.

Area cancellation is not a local test



Module 4 reference illustration: a signed integral can hide compensating local errors.

Different tests answer different questions

DIAGNOSTIC	EVIDENCE AND LIMITATION
Area	Global signed integral; endpoint coverage matters
Point/local	Local deviations; differentiation amplifies noise
Model-based	Deviations relative to the chosen model
Herington	Empirical temperature-range correction

Agreement across tests strengthens a diagnosis but does not prove the data are correct.

Isobaric data need temperature information

At approximately constant P,

$$\sum_i x_i d \ln \gamma_i = -\frac{h^E}{RT^2} dT$$

Independent excess-enthalpy data can support the temperature correction. An empirical range correction does not replace measured h^E .

Missing endpoints require an explicit extrapolation assumption.

Fitting objective and parameter convention

Lab 08 minimizes the mean of

$$\left(100 \frac{P_{pred} - P_{obs}}{P_{obs}}\right)^2 + [100(y_{pred} - y_{obs})]^2$$

Compare one-parameter Margules, two-parameter Margules and NRTL on the same rows. NRTL α is fixed. This objective has no measurement-uncertainty weighting.

Residual structure matters

A lower scalar error may accompany biased residuals near an endpoint or a narrow temperature interval.

Compare residuals against composition and temperature. Report parameter bounds and temperature convention.

Lab 08 compares three models on one fixed held-out split. Its pairs bootstrap reports conditional fit spread, not a new-measurement prediction interval.

Class demonstration • Labs 03 and 08

Consistency diagnostics • Student VLE input

1. Compare a clean dataset with a controlled defect in Lab 03.
2. Load Lab 08's synthetic example and inspect its Antoine convention.
3. Compare held-out errors on a fixed split; inspect conditional bootstrap spread.

Synthetic examples are training data, not experimental validation.

Guided exploration • 35 minutes

Use your dataset or the provided synthetic example.

Record source, component order, units and valid range. Compare two models. Explain one residual pattern and one unsupported extrapolation.

Save the study JSON, residual CSV and worksheet. If real data are unavailable, clearly label the synthetic case.

A fitted model has a domain

A single-temperature fit supplies composition dependence at that temperature. It does not establish $d\gamma/dT$.

A VLE fit need not identify LLE accurately. Similar VLE residuals can conceal different liquid Gibbs-energy curvature.

Session 3 adds stability and liquid coexistence evidence.

Exit question

A dataset passes an area criterion and NRTL gives a small pressure error.

Give two reasons why this is insufficient to claim accurate LLE predictions.

What additional observation or check would reduce each uncertainty?

Independent practice • suggested 60–90 minutes

Assessment set 2 and rubric • [Offline package](#)

One dataset, at least two activity models, residuals and a justified validity statement.

Retain the calculator export, your worksheet, a comparison plot/table and one independent check.

State an assumption that limits your conclusion.

Use the core labs on the [learning path](#). Optional extensions are additional work.

References and further study

[Module reference deck](#) · [Lab sources and equations](#)

Derivations and original figure references remain in the corresponding module deck. Each lab records its implemented equations and assumptions.

Synthetic worked examples illustrate calculations; they are not evidence of real-system accuracy.